

Unified constraint framework for exotic spin-dependent interactions: a matter–antimatter sector comparison

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Exotic spin-dependent interactions mediated by light, weakly coupled bosons are described in two disjoint languages: the Standard Model Extension (SME), a relativistic effective field theory of Lorentz and CPT violation, and the Dobrescu–Mocioiu (DM) classification, a complete basis of sixteen non-relativistic potentials V_1 – V_{16} in which experiments report their bounds. We construct a systematic dictionary between the two by applying the Foldy–Wouthuysen transformation to three SME fermion-sector coefficients—the CPT-odd vector b_μ and the CPT-even tensors $H_{\mu\nu}$ and $d_{\mu\nu}$ —and matching the resulting non-relativistic Hamiltonians onto the DM basis. All three reductions are obtained in full and agree with the literature, including the momentum-linear d_{ij} and d_{00} components, whose apparent sign discrepancy we trace to a momentum-index convention in the reference result rather than to the derivation. We then define a dimensionless CPT asymmetry parameter $A_\alpha = (g_\alpha^f - g_\alpha^{\bar{f}})/(g_\alpha^f + g_\alpha^{\bar{f}})$ and evaluate it against a compiled database of 247 experimental constraint datasets standardized to a common interaction-range axis. Fifteen matter–antimatter pairs can be compared, and all exhibit statistically significant asymmetry after correction for autocorrelation among interpolated points. We show analytically, and confirm across all fifteen pairs, that A_α constructed from one-sided experimental upper bounds cannot by itself distinguish CPT violation from an ordinary sensitivity gap between the two experiments: a CPT-even channel in our sample produces an asymmetry comparable to the CPT-odd channels. The binding constraint on this class of CPT test is therefore the absence of antimatter-sector measurements rather than statistical power. Only 19 of 247 analyzed datasets involve an antimatter sector, and seven of twelve classified potentials have none at all. A coverage analysis identifies the e - N , n - N and p - N sectors and the V_{1a} , V_{9+10} and V_{12+13} potentials as the largest opportunities for new measurement. The mapping table, database and analysis pipeline are released as an extensible resource.

I. INTRODUCTION

The Standard Model is tested to extraordinary precision yet remains incomplete: it does not accommodate quantum gravity, offers no dark matter candidate, and does not explain the observed baryon asymmetry [1]. A well-motivated class of extensions introduces light, weakly coupled bosonic fields—axions, axion-like particles, dark photons—that mediate feeble long-range forces, a possibility first set out systematically by Moody and Wilczek [10]. Those that couple to fermion spin rather than to charge or mass alone are of particular interest, since spin has no classical analogue and therefore offers an unusually clean channel for probing symmetry violation.

Two formalisms describe such interactions, and they are not directly commensurable. The SME of Colladay and Kostelecký [2, 3], motivated by the spontaneous Lorentz breaking identified in string theory [4], is a relativistic effective field theory that parametrizes Lorentz and CPT violation through fixed background coefficients such as b_μ , $H_{\mu\nu}$ and $d_{\mu\nu}$; its coefficients are catalogued and constrained in the data tables of Kostelecký and Russell [7]. The DM classification [9] is a complete non-relativistic basis of sixteen potentials V_1 – V_{16} , generated

by single-boson exchange between two fermions and recast in coordinate space for laboratory use in Ref. [21], in which essentially all laboratory searches report their limits; the experimental landscape through 2024 is reviewed comprehensively in Ref. [12]. Translating between the two requires the Foldy–Wouthuysen (FW) transformation [11], the standard technique for extracting the non-relativistic limit of a relativistic Dirac Hamiltonian.

Separately, antimatter precision physics has become one of the sharpest available probes of fundamental symmetry. Experiments at CERN compare matter and antimatter magnetic moments and spectral lines at a level that directly tests CPT invariance [14, 15]. Joining these two threads suggests a concrete observable: a CPT-violating spin-dependent interaction should appear as an asymmetry between the coupling constants inferred for matter and for antimatter.

No complete dictionary between the SME and DM bases has been published, and no framework exists for using such a dictionary to compare matter-sector and antimatter-sector constraints within a single quantitative measure. Prior mappings are partial, restricted to subsets of coefficients or potentials, and the constraint literature itself is reported in inconsistent units and conventions, which obstructs direct comparison across platforms. This paper supplies both the dictionary and the comparison framework, and reports what the currently published data can and—importantly—cannot establish.

Section II presents the FW reductions and the result-

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ing SME→DM matching table. Section III defines the asymmetry parameter and derives the central interpretive limitation on it. Section IV describes the compiled database and the statistical methodology. Sections V and VI report the matched-pair results and the coverage analysis. Section VII states the limitations of the present database, and Sec. VIII concludes.

II. SME-TO-DM MAPPING

A. The b_μ coefficient

The CPT-odd term $\mathcal{L}_b = -b_\mu \bar{\psi} \gamma_5 \gamma^\mu \psi$ separates into a spatial part b_i , which is even and therefore exact at order m^0 , and a temporal part b_0 , which is odd and enters only at order $1/m$. In the Dirac representation $\gamma^0 \gamma_5 \gamma^i = -\Sigma^i$, so projected onto the upper block

$$H_{\text{NR}}(b_i) = -\mathbf{b} \cdot \boldsymbol{\sigma} \quad (\text{matter}), \quad (1)$$

already in non-relativistic form with no further FW iteration required. The temporal component is odd, since $\gamma^0 \gamma_5 \gamma^0 = -\gamma_5$, and so contributes through the FW cross-term between the free odd operator $\boldsymbol{\alpha} \cdot \mathbf{p}$ and the perturbing odd operator $\mathcal{O}_{b_0} = -b_0 \gamma_5$. Retaining only the piece linear in b_0 ,

$$H_{\text{NR}}(b_0) = \frac{\beta}{2m} \{ \boldsymbol{\alpha} \cdot \mathbf{p}, \mathcal{O}_{b_0} \} \Big|_{\text{upper}} = -\frac{b_0(\boldsymbol{\sigma} \cdot \mathbf{p})}{m} + O(m^{-2}). \quad (2)$$

The antiparticle sector follows from the block structure of $\gamma^0 \gamma_5 \gamma^i = -\Sigma^i$ rather than from a charge-conjugation transformation of the Lagrangian. Since $\Sigma^i = \text{diag}(\sigma^i, \sigma^i)$ repeats the *same* operator in the upper ($\beta = +1$) and lower ($\beta = -1$) blocks,

$$\langle u | b_i \gamma^0 \gamma_5 \gamma^i | u \rangle = \langle l | b_i \gamma^0 \gamma_5 \gamma^i | l \rangle = -b_i \sigma^i. \quad (3)$$

By the hole-theory prescription a negative-energy solution ϕ_- with $H\phi_- = -|E|\phi_-$ is reinterpreted as a positive-energy antiparticle through an overall sign flip of the Hamiltonian, since $(-H)\phi_- = +|E|\phi_-$. The antiparticle therefore sees every perturbation with the sign opposite to its lower-block value in Eq. (3):

$$H_{\text{NR}}^{\bar{f}}(b_i) = -(-\mathbf{b} \cdot \boldsymbol{\sigma}) = +\mathbf{b} \cdot \boldsymbol{\sigma}. \quad (4)$$

This is a genuine matter–antimatter sign flip, the qualitative signature of a CPT-odd coefficient. It is worth emphasizing that it is *not* a charge-conjugation argument on the axial-vector bilinear $\bar{\psi} \gamma_5 \gamma^\mu \psi$, which is C -even for every μ and therefore cannot by itself produce the difference between Eqs. (1) and (4); the flip originates specifically in the hole-theory reinterpretation of the negative-energy solution. This is consistent with the charge-conjugation treatment of Bluhm, Kostelecký and Russell [8], whose substitution rule taking hydrogen’s SME couplings to antihydrogen’s flips a_μ , $d_{\mu\nu}$ and $H_{\mu\nu}$ but not b_μ ; their

eventual sign flip in the observable frequency shift arises instead from the opposite spin-to-sublevel assignment in the bound antiatom, a bound-state mechanism distinct from, but pointing to the same conclusion as, the free-particle argument used here.

Promoting Eq. (1) to a two-body potential by pairing the same vertex on both fermions gives the spin–spin structure

$$V_2(r) = -g_A^{(1)} g_A^{(2)} \frac{1}{4\pi} (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) \frac{e^{-r/\lambda}}{r}, \quad (5)$$

identifying b_i with V_2 at leading order, while b_0 matches the velocity-dependent V_8 at order $1/m$.

B. The $H_{\mu\nu}$ coefficient

$H_{\mu\nu}$ is CPT-even but Lorentz-odd: matter and anti-matter couple identically at tree level, so its experimental signature is a sidereal anisotropy rather than a matter–antimatter asymmetry. Its magnetic-like spatial components H_{ij} are even and generate, at order m^0 , using $\gamma^0 \sigma^{ij} \Big|_{\text{upper}} = \varepsilon^{ijk} \sigma^k$,

$$H_{\text{NR}}(H_{ij}) = +\mathcal{H}_B \cdot \boldsymbol{\sigma}, \quad \mathcal{H}_B^k = \frac{1}{2} \varepsilon^{ijk} H_{ij}, \quad (6)$$

matching the dipole–dipole potential V_3 . The electric-like mixed components H_{0i} are odd, since $\gamma^0 \sigma^{0i} = i\gamma^i$ is block-off-diagonal, and generate at order $1/m$, through the same FW cross-term used for b_0 with $\mathcal{O}_{H_E} = i \sum_k H_{0k} \gamma^k$,

$$H_{\text{NR}}(H_{0i}) = +\frac{1}{m} \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E), \quad H_E^i \equiv H_{0i}, \quad (7)$$

matching the spin–velocity potential V_7 . Both components being CPT-even, $H_{\text{NR}}^f = H_{\text{NR}}^{\bar{f}}$ in each case, so this coefficient predicts $A_\alpha \approx 0$ for a comparably sensitive matter–antimatter pair, in contrast to b_μ .

C. The $d_{\mu\nu}$ coefficient

Following Kostelecký and Lane [5], $d_{\mu\nu}$ is a CPT-even, traceless, kinetic-sector coefficient entering through the derivative coupling $\mathcal{L}_d = \frac{1}{2} i \bar{\psi} d^{\mu\nu} \gamma_5 \gamma_\mu \overleftrightarrow{\partial}_\nu \psi$. With $\Gamma_\mu \equiv \gamma^0 \gamma_5 \gamma_\mu$ one finds $\Gamma_0 = -\gamma_5$ (odd) and $\Gamma_i = +\Sigma^i$ (even), from which

$$H_{\text{NR}}(d_{i0}) = +d_{i0} m \sigma^i \rightarrow V_2, \quad (8)$$

$$H_{\text{NR}}(d_{ij}) = +d_{ij} p^j \sigma^i \rightarrow V_7, V_8, \quad (9)$$

$$H_{\text{NR}}(d_{00}) = +d_{00} (\boldsymbol{\sigma} \cdot \mathbf{p}) \rightarrow V_8. \quad (10)$$

All three results reproduce the non-relativistic Hamiltonian of Ref. [5] exactly, including signs, once that reference’s momentum convention is applied. The convention is not stated explicitly alongside their Eq. (4), which describes p_j as the three-momentum of the particle, but it

is fixed unambiguously by the free-particle term of the same authors' intermediate relativistic Hamiltonian [6], $m\mathcal{P}_0 := -p_j\gamma^0\gamma^j$. That expression equals $-\boldsymbol{\alpha} \cdot \mathbf{p}$, so recovering the free Dirac Hamiltonian $+\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$ requires their lower-index p_j to be the covariant component $p_j = -p^j$. Read as the physical three-momentum instead, the two momentum-linear terms appear to differ by an overall sign while the mass-enhanced term still agrees—the signature of a momentum-index mismatch rather than of a derivation error. We have verified the correspondence symbolically, including an independent implementation of the kinetic-sector field redefinition $A = 1 - \frac{1}{2}\gamma^0(\Gamma_0 - \gamma_0)$ of Ref. [6].

D. Master matching table

Table I synthesizes the three reductions. A caveat on its final column applies uniformly to every row: it records the naive signature *consistent with* the coefficient's CPT parity, not a claim that the parity *causes* an observed asymmetry, for the reason derived in Sec. III.

III. THE CPT ASYMMETRY PARAMETER

For a coupling α constrained independently in a matter sector and its conjugate antimatter sector, we define the dimensionless asymmetry

$$A_\alpha(\lambda) = \frac{g_\alpha^f(\lambda) - g_\alpha^{\bar{f}}(\lambda)}{g_\alpha^f(\lambda) + g_\alpha^{\bar{f}}(\lambda)}, \quad (11)$$

evaluated as a function of interaction range λ . Exact CPT invariance with comparably sensitive experiments gives $A_\alpha = 0$.

It is tempting to substitute the signed relation $g_{\bar{f}} = -g_f$ implied by Eqs. (1) and (4) directly into Eq. (11). Doing so gives

$$A_\alpha = \frac{g - (-g)}{g + (-g)} = \frac{2g}{0} \longrightarrow \infty, \quad (12)$$

a genuine divergence—confirmed by symbolic limit calculation—rather than a value saturating at unity.

This is not, however, the quantity that published data can supply. The inputs g_f and $g_{\bar{f}}$ are independent, strictly positive experimental *upper bounds*, not signed measurements of one underlying coupling. For any two independent positive bounds g_1, g_2 the ratio $(g_1 - g_2)/(g_1 + g_2)$ approaches ± 1 whenever one bound is far tighter than the other, *regardless of the underlying physics*: setting $g_{\text{tight}} = r g_{\text{loose}}$ and taking $r \rightarrow 0^+$ gives $A_\alpha \rightarrow 1$ with no CPT content whatever.

Consequently an observed $|A_\alpha| \approx 1$ is *consistent with*, but is not *evidence for*, a non-zero b_μ : the same numerical pattern is exactly what independent bounds of very different sensitivity produce under exact CPT symmetry.

This governs the interpretation of every asymmetry reported below, and we regard it as a central result rather than an incidental caveat. Its practical consequence is that the most informative pairs are those with the *lowest* asymmetry, since those are the pairs in which the two sectors are probed at comparable sensitivity.

IV. DATABASE AND METHODS

A. Compilation and standardization

We compiled 283 experimental constraint datasets from the published literature, drawing on the compilation of Ref. [12] and the underlying primary sources, spanning nitrogen-vacancy-center magnetometry [27, 31, 32], torsion balances [25, 26, 28, 30, 34], atomic co-magnetometers [35], precision spectroscopy of simple atoms and molecules [16, 17, 23, 33], Casimir-force measurements [18, 19], and antimatter experiments [20, 24]. Each dataset is a digitized exclusion curve $g(\lambda)$ or $\alpha(\lambda)$.

Datasets are parsed for coupling family, DM potential label, fermion sector, and source. A bare numeric prefix in the source filenames denotes the potential number itself, so a 3-prefixed file bounds V_3 alone and only a 23 prefix denotes the $V_2 + V_3$ combination. Every potential assignment was validated against the source compilation's own curation records; among the 194 datasets those records cover there is no disagreement. Interaction ranges reported variously in meters, MeV^{-1} , eV^{-1} , centimeters, nanometers and femtometers are converted to a common meter scale by a unit-detection routine applied before any downstream analysis. Bounds published as a gravity-normalized Yukawa strength $|\alpha|$ are kept separate from those published as a coupling $|g|$, the two differing by roughly 37 decades.

B. Exclusions

Of the 283 compiled datasets, 36 are held out of the analysis, leaving 247. Twenty-eight are combined envelope curves, twelve in $g_A g_V$ and sixteen in $g_p g_s$, that aggregate several experiments; the source compilation assigns them no single potential and flags them for review, and no single V_n applies to an envelope by construction. One is an astrophysical $e-N$ bound with no potential assignment that can be established from the source. Two are contributor-template placeholders. Five are the 90% confidence-level variants of the Cong *et al.* hydrogen bounds [13], which the source ships alongside the 95% CL curves; they are the same measurement at a second confidence level, so retaining both would double-count the experiment. Every exclusion is recorded individually with a stated reason, and the registry retains all 273 rows so that the compiled and analyzed counts remain separately auditable.

TABLE I. Master SME→DM matching table. Every entry in the final column is a naive signature only—consistent with, not caused by, the stated CPT parity. A sensitivity gap between independent one-sided bounds reproduces the same pattern regardless of true CPT status (Sec. III).

SME coeff.	CPT	Non-relativistic Hamiltonian	DM potential(s)	1/ m order	A_α signature
b_i	Odd	$-\mathbf{b} \cdot \boldsymbol{\sigma}$ (matter); $+\mathbf{b} \cdot \boldsymbol{\sigma}$ (antimatter)	V_2	m^0	$ A_\alpha \rightarrow 1$ predicted
b_0	Odd	$-b_0(\boldsymbol{\sigma} \cdot \mathbf{p})/m$	V_8	m^{-1}	$ A_\alpha \rightarrow 1$ predicted
H_{ij}	Even	$+\mathcal{H}_B \cdot \boldsymbol{\sigma}$	V_3	m^0	$A_\alpha \approx 0$ if comparably sensitive
H_{0i}	Even	$+\frac{1}{m}\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E)$	V_7	m^{-1}	$A_\alpha \approx 0$ if comparably sensitive
d_{i0}	Even	$+d_{i0} m \sigma^i$	V_2	m^{+1}	$A_\alpha \approx 0$; derived and verified
d_{ij}	Even	$+d_{ij} p^j \sigma^i$	V_7, V_8	m^0 (vel.)	$A_\alpha \approx 0$; derived and verified
d_{00}	Even	$+d_{00}(\boldsymbol{\sigma} \cdot \mathbf{p})$	V_8	m^0 (mom.)	$A_\alpha \approx 0$; derived and verified

C. Pairing and statistics

Two datasets are matched into a matter–antimatter pair only when they share coupling family, DM potential label and interaction class, their fermion sectors are physically conjugate (for example ee with $e\bar{e}$), and exactly one is flagged as antimatter. This logic is deliberately conservative: it will not match datasets differing in potential classification even where their sectors are conjugate.

For each matched pair both curves are interpolated logarithmically in $\log \lambda$ – $\log g$ onto a common log-spaced grid of 300 points spanning the overlapping λ range, and $A_\alpha(\lambda)$ is evaluated at every point. Significance is assessed in four stages, each addressing a limitation of the previous one:

1. A uniform χ^2 test assuming a flat 10% fractional uncertainty at every grid point.
2. A weighted χ^2 test estimating a per-point fractional uncertainty from the local curvature of each curve in log–log space, on the grounds that a smoothly varying published bound is better determined than one with sharp local features.
3. Both tests above treat the 300 grid points as independent, which is unjustified: they are interpolated from far fewer real measurements along smooth curves, so adjacent points are strongly correlated. An effective number of degrees of freedom is estimated from the autocorrelation length of the χ^2 residuals in grid-index space (the lag at which the residual autocorrelation first falls below $1/e$), and the weighted significance is recomputed against it.
4. A parametric bootstrap (2000 resamples) gives a 95% confidence interval on the mean $|A_\alpha|$.

Two properties of the bootstrap deserve explicit statement. It is a delta-method approximation rather than a resampling of raw coupling values: by that stage each pair has been reduced to $A_\alpha(\lambda)$ and its combined uncertainty, so each resample perturbs a linearization of Eq. (11) about $g_m \approx g_a$. And because the interval is computed on $|A_\alpha|$ rather than the signed asymmetry, it

is systematically biased away from zero: folding a distribution centered at zero through an absolute value can only push resampled values upward. A bootstrap interval excluding zero should therefore not, on its own, be read as evidence of non-zero asymmetry.

All analysis is performed by SPINDEP, an open, version-controlled pipeline; the full analysis is reproducible from a single command, and the parsing, matching and statistical modules are covered by an automated test suite.

V. RESULTS

A. The compiled database

The 247 analyzed datasets divide by coupling family as g_{AgA} (66), $g_s g_s$ (49), $g_V g_V$ (44), $g_p g_s$ (37), g_{AgV} (25), $g_p g_p$ (15) and an unlabeled V_1 family (11). By DM potential the database is dominated by V_{4+5} (61), followed by V_1 (47), V_3 (39), V_{9+10} (31), V_2 (20) and V_{12+13} (16), with V_{1a} (14), V_{11} (7), V_8 (6), V_{2+3} (2), V_{16} (2) and V_{15} (2) making up the remainder. Several labels combine adjacent catalogue potentials that the underlying experiments do not distinguish; V_6 , V_7 and V_{14} have no classified dataset at all. Figure 1 shows the constraint atlas for V_3 , the best-covered potential in the antimatter sector.

Only 19 of the 247 analyzed datasets involve an antimatter sector: 9 in $e\bar{p}$, 5 in $e\bar{e}^+$, 3 in $e\bar{\mu}$, and one each in $\bar{p}$ -He and the $dd\mu^+$ molecular ion. This asymmetry—19 antimatter datasets against 228 matter-sector datasets—is the central empirical fact of this work.

B. Matched pairs

The pairing logic returns fifteen matter–antimatter pairs, listed in Table II with the mean absolute asymmetry, the autocorrelation-corrected effective degrees of freedom, the resulting significance, and the 95% bootstrap interval.

Effective degrees of freedom range from 6 to 21 across the fifteen pairs, a factor of 14–50 below the naive 300

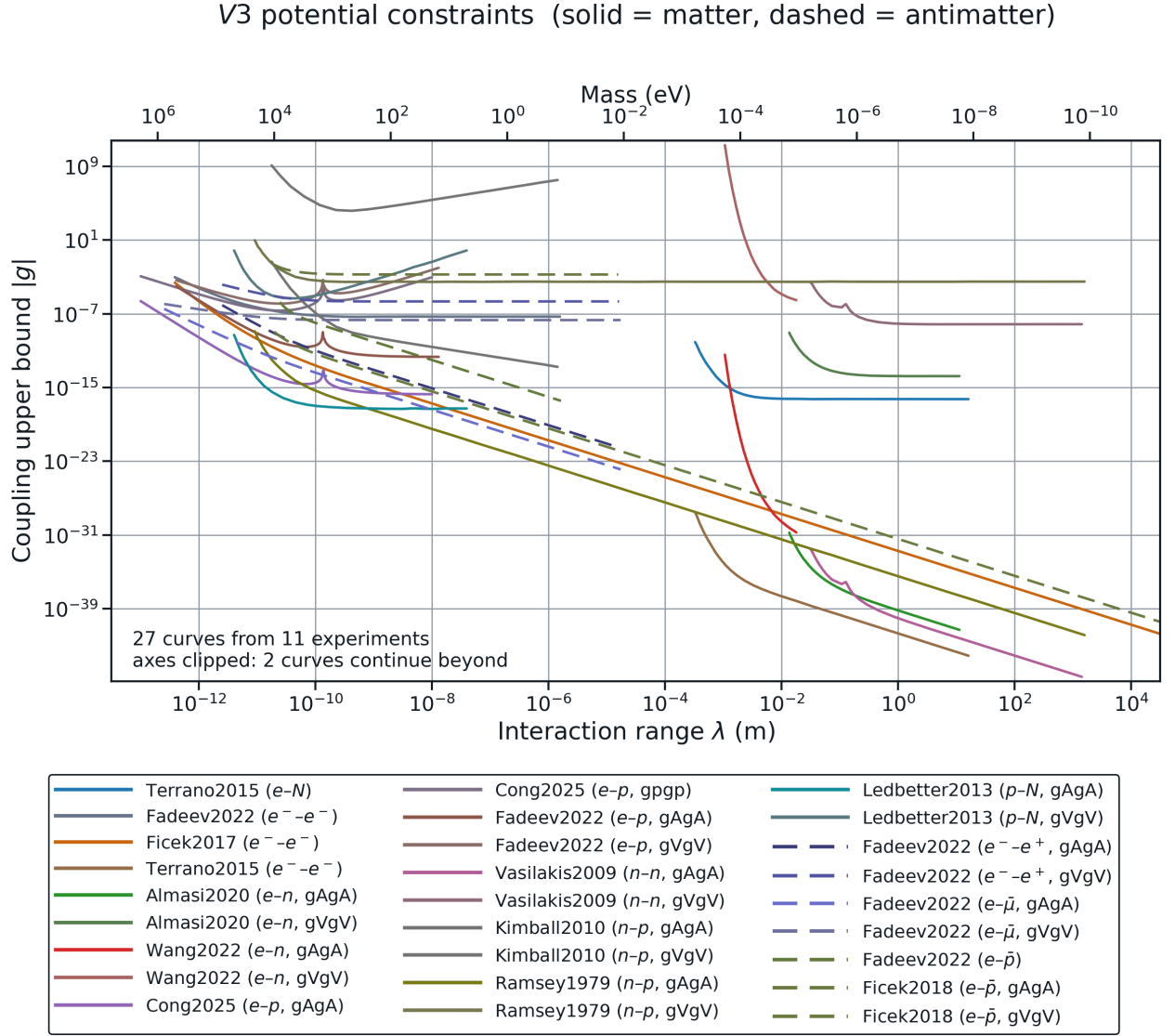


FIG. 1. Constraint atlas for V_3 , the potential with the broadest antimatter-sector coverage in the compiled database. Solid curves are matter-sector bounds, dashed curves antimatter-sector; the region above each curve is excluded. Shown at full page width because the panel carries 25 curves from 10 experiments.

grid points, confirming that the autocorrelation correction is not a negligible refinement. Every pair remains significant at effectively $p \rightarrow 0$ after correction.

C. Interpretation

By Sec. III, an asymmetry near unity in Table II is consistent with, but not evidence for, CPT violation: a large sensitivity gap between the two measurements produces the same signature regardless of the underlying physics. This applies uniformly to every row, including the $g_{AgA} \cdot V_2 \cdot e-\bar{p}$ pair at 0.9998, the highest in the database.

The $g_s g_s \cdot V_1 \cdot e-e^+$ pair at 0.8727 makes the point concretely. Its scalar-scalar coupling is CPT-even at lead-

ing order, so its asymmetry is attributable entirely to a sensitivity gap. That a CPT-even channel produces an asymmetry comparable to the CPT-odd g_{AgA} channels is a useful internal check on the argument of Sec. III: high asymmetry alone does not discriminate CPT-odd physics from CPT-even physics measured with mismatched sensitivity. Figures 2 and 3 contrast the highest- and lowest-asymmetry pairs.

The lowest entry, 0.3336 for the $g_{AgA} \cdot V_2 \cdot e-e^+$ pair of Refs. [27, 29], is the pair whose two bounds are closest in sensitivity, and is therefore the single most informative datum currently available for constraining b_μ in the $e-e^+$ channel—precisely because it is not saturated.

The four pairs contributed by the recent hydrogen-spectroscopy bounds of Ref. [13] make the same point from the opposite direction. Those bounds are substan-

TABLE II. All fifteen matched matter–antimatter pairs, with the autocorrelation-corrected effective degrees of freedom dof_{eff} , the resulting weighted- χ^2 significance p_{eff} , and the 95% bootstrap confidence interval on the mean asymmetry. Per Sec. III, a value near unity is consistent with, but not evidence for, CPT violation.

Coupling	Potential	Sector	Matter / antimatter source	$ A_\alpha $	dof_{eff}	95% CI
g_{AgA}	V_3	$e\text{-}\bar{p}$	Cong [13] / Fadeev [22]	1.0000	8	[0.9995, 0.9997]
g_{AgA}	V_2	$e\text{-}\bar{p}$	Cong [13] / Ficek [24]	0.9999	21	[0.9995, 0.9996]
g_{AgA}	V_2	$e\text{-}\bar{p}$	Karshenboim [29] / Ficek [24]	0.9998	17	[0.9994, 0.9996]
g_{AgA}	V_2	$e\text{-}e^+$	Ficek [23] / Karshenboim [29]	0.9892	21	[0.9891, 0.9893]
g_{AgA}	V_3	$e\text{-}e^+$	Ficek [23] / Fadeev [22]	0.9539	6	[0.9538, 0.9540]
g_{VgV}	V_3	$e\text{-}e^+$	Fadeev [22] / Fadeev [22]	0.9535	6	[0.9534, 0.9537]
g_{pgp}	V_3	$e\text{-}e^+$	Fadeev [22] / Fadeev [22]	0.9535	6	[0.9534, 0.9537]
g_{pgp}	V_3	$e\text{-}\bar{p}$	Cong [13] / Ficek [24]	0.9304	6	[0.9302, 0.9304]
g_{AgA}	V_3	$e\text{-}\bar{p}$	Cong [13] / Ficek [24]	0.9303	6	[0.9301, 0.9303]
g_{sgs}	V_1	$e\text{-}e^+$	Delaunay [20] / Adkins [16]	0.8727	21	[0.8726, 0.8728]
g_{AgA}	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Ficek [24]	0.8236	6	[0.8235, 0.8237]
g_{AgA}	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Fadeev [22]	0.8044	8	[0.8043, 0.8045]
g_{VgV}	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Ficek [24]	0.7994	7	[0.7993, 0.7995]
g_{pgp}	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Ficek [24]	0.7994	7	[0.7993, 0.7995]
g_{AgA}	V_2	$e\text{-}e^+$	Jiao [27] / Karshenboim [29]	0.3336	9	[0.3335, 0.3337]

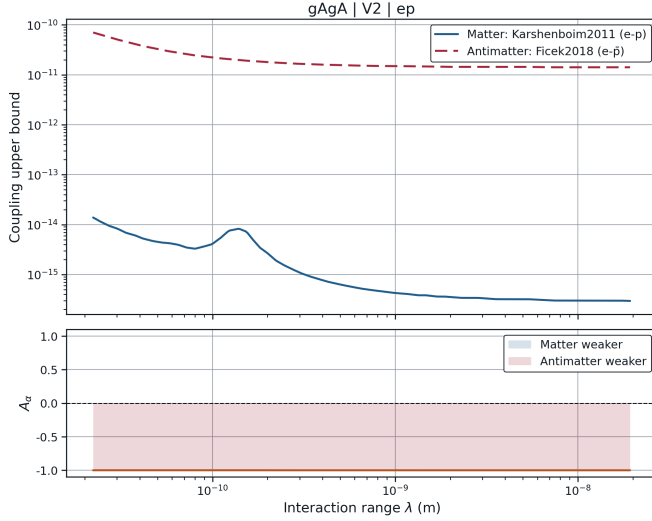


FIG. 2. Matter–antimatter comparison for the $g_{AgA} \cdot V_2 \cdot e\text{-}\bar{p}$ pair, the highest asymmetry in the database ($|A_\alpha| = 0.9998$). The two bounds differ by orders of magnitude in sensitivity over the overlapping range, which is sufficient on its own to saturate A_α .

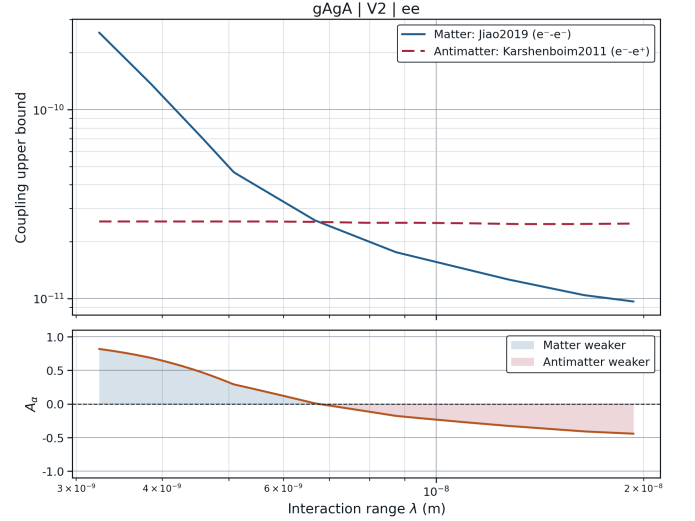


FIG. 3. Matter–antimatter comparison for the $g_{AgA} \cdot V_2 \cdot e\text{-}e^+$ pair, the lowest asymmetry in the database ($|A_\alpha| = 0.3336$). The two sectors are probed at comparable sensitivity here, making this the most informative pair in the present database despite its smaller asymmetry.

VI. COVERAGE GAP ANALYSIS

A. By fermion sector

tially tighter on the matter side than any previously available $e\text{-}p$ constraint, and every pair they enter sits at the top of Table II, the $g_{AgA} \cdot V_3$ pair reaching 1.0000 to four decimal places. Improving the matter-sector measurement, with no change to the antimatter sector and none to the underlying physics, drove the asymmetry *up*. This is a direct empirical confirmation of the argument of Sec. III: A_α built from one-sided bounds tracks the sensitivity ratio, not CPT status.

Table III compares, for each of the eleven canonical matter–antimatter *sector* pairs, the number of analyzed datasets on each side. These are pairings of fermion sectors, not the eleven matched dataset pairs of Sec. V; the two counts coincide at eleven by coincidence.

The three nucleus sectors dominate the gap. $e\text{-}N$ is the single largest opportunity, with 54 matter-sector datasets against zero in $e\text{-}\bar{N}$, but $n\text{-}N$ is barely behind at 49 and $p\text{-}$

TABLE III. Matter versus antimatter dataset counts per canonical fermion sector pair. ^aA $\bar{p}$ - N bound does exist—the antiprotonic-helium spectroscopy of Ref. [24]—but is published as a gravity-normalized strength $|\alpha|$ rather than a coupling $|g|$, so it cannot be compared with the p - N datasets without a conversion not performed here. The gap in *comparable* data is real, but it is a unit-conversion gap rather than an absence of measurement.

Matter	Antimatter	Matter	Anti	Status
$e-e$	$e-e^+$	40	5	Populated
$e-p$	$e-\bar{p}$	17	9	Populated
$e-\mu$	$e-\bar{\mu}$	3	3	Populated, unmatched
$e-n$	$e-\bar{n}$	16	0	Gap
$\mu-\mu$	$\mu-\bar{\mu}$	1	0	Gap (thin both sides)
$n-p$	$n-\bar{p}$	9	0	Gap
$n-n$	$n-\bar{n}$	8	0	Gap
$p-p$	$p-\bar{p}$	0	0	No data either side
$e-N$	$e-\bar{N}$	50	0	Largest gap
$n-N$	$n-\bar{N}$	49	0	Gap (second largest)
$p-N$	$p-\bar{N}$	18	0 ^a	Gap in comparable units

N adds a further 18; between them these three account for 117 of the 228 matter-sector datasets and carry no antimatter-sector counterpart at all. The $e-n$, $n-p$ and $n-n$ sectors each have a smaller but still substantial base (8–16 datasets) with no counterpart. Even the $e-\mu/e-\bar{\mu}$ pair, populated on both sides, has not been matched by the pairing logic, which merits direct investigation.

The eleven sector pairs account for 211 matter-sector and 17 antimatter-sector datasets, 228 in all, of the 247 analyzed. The remaining 19 do not belong to a matter–antimatter sector pair: 16 are astrophysical bounds quoted for a single sector with no laboratory counterpart, and three are the μ - N , $\bar{p}$ -He and $dd\mu^+$ entries, whose sectors have no counterpart row. The last two of these are also why the antimatter column totals 17 rather than 19.

B. By potential

Table IV breaks the same comparison down by DM potential. Seven of the twelve classified potentials— V_{11} , V_{15} , V_{16} , V_{1a} , V_{9+10} , V_{12+13} and V_{2+3} —have *zero* antimatter-sector datasets. Only V_3 has antimatter coverage exceeding a handful, which is why fourteen of the fifteen matched pairs involve V_2 or V_3 .

Figures 4 and 5 display the same structure graphically. Both count *independent bounds* rather than dataset files: 48 of the 247 analyzed datasets are the same measured bound filed under more than one coupling family, and are collapsed to a single curve before plotting, leaving 199 independent bounds. Totals read off these figures are therefore systematically smaller than the counts in Tables III and IV, which count files. The two views agree on every gap; they differ only in how multiply-filed bounds are counted.

TABLE IV. Matter versus antimatter dataset counts per Dobrescu–Mocioiu potential.

Potential	Matter	Anti	Status
V_{4+5}	58	3	Thin antimatter coverage
V_{9+10}	31	0	Gap
V_3	29	10	Best covered
V_{1a}	14	0	Gap
V_2	18	2	Thin antimatter coverage
V_1	44	3	Thin antimatter coverage
V_{12+13}	16	0	Gap
V_{11}	7	0	Gap
V_8	5	1	Thin antimatter coverage
V_{15}	2	0	Gap
V_{16}	2	0	Gap
V_{2+3}	2	0	Gap (combination label)

C. Proposed experimental priorities

Three directions follow. First, antimatter-sector effort is best prioritized in the $e-N$, $n-N$, $p-N$, $e-n$, $n-p$ and $n-n$ sectors, which together account for 150 of the 228 matter-sector datasets and zero antimatter-sector ones; precision antiproton experiments such as BASE [14] and antihydrogen spectroscopy such as ALPHA [15] are the most direct existing platforms. Second, the same logic applies at the potential level to V_{1a} , V_{9+10} and V_{12+13} , each with 16–31 matter-sector datasets and none on the antimatter side; V_1 , V_2 and V_8 have thin but non-zero coverage and are better placed as candidates for improving existing precision than for opening a channel from zero. Third, following the precedent of Ref. [12], in which existing co-magnetometer data collected for sidereal Lorentz-violation searches was reinterpreted as bounds on several V_n terms, a systematic reanalysis of existing BASE, ALPHA and co-magnetometer data is likely to close part of the gap faster, and at lower cost, than commissioning new measurements.

Across all three, the asymmetry argument of Sec. III implies a specific design criterion: a new antimatter-sector bound is scientifically useful for a CPT test in proportion to how closely it *matches* the sensitivity of the corresponding matter-sector bound, not how tight it is in absolute terms. A bound many decades looser than its matter-sector counterpart will saturate A_α and carry no CPT information.

VII. LIMITATIONS

Two limitations of the present database are distinct from the physics coverage gaps above.

First, the 36 datasets held out of the analysis are excluded rather than reclassified, and the reasons differ in kind. Twenty-eight are combined envelope curves, twelve in g_{AgV} and sixteen in $g_p g_s$, that the source compilation marks for review and assigns no potential; these are

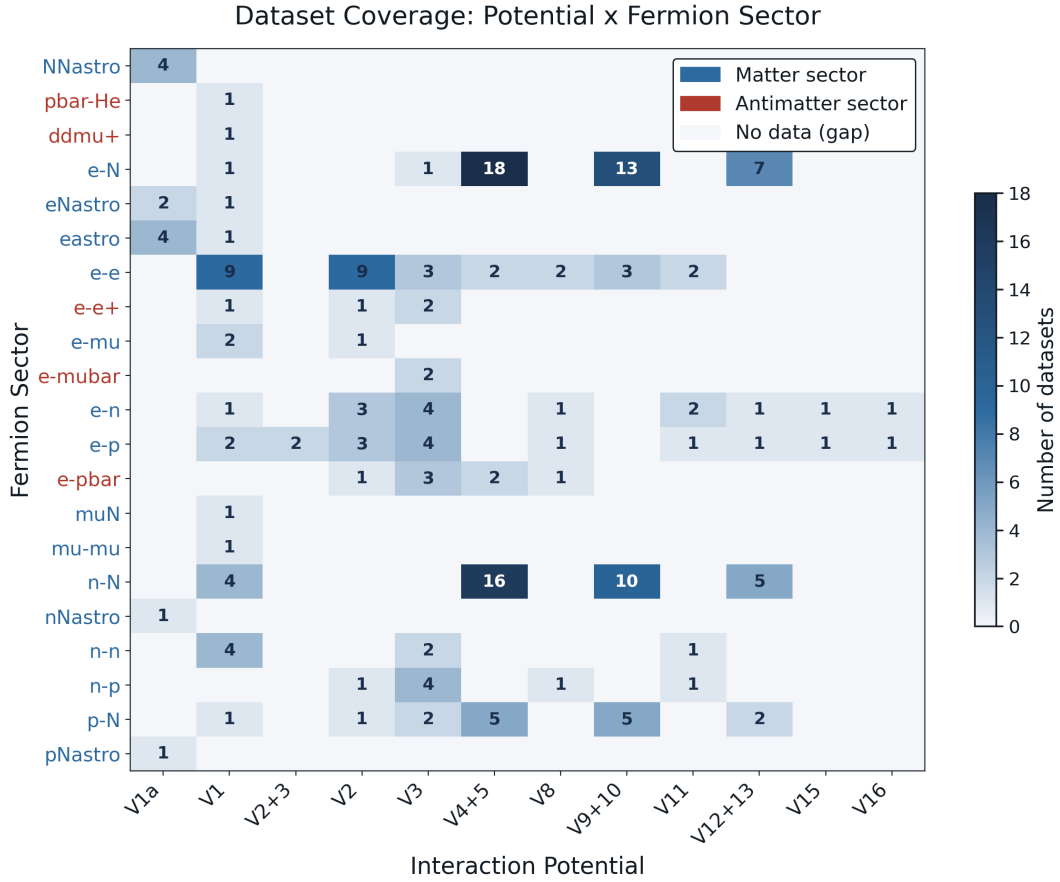


FIG. 4. Dataset coverage by potential and fermion sector; antimatter-sector rows are highlighted. Cells count independent bounds, not dataset files.

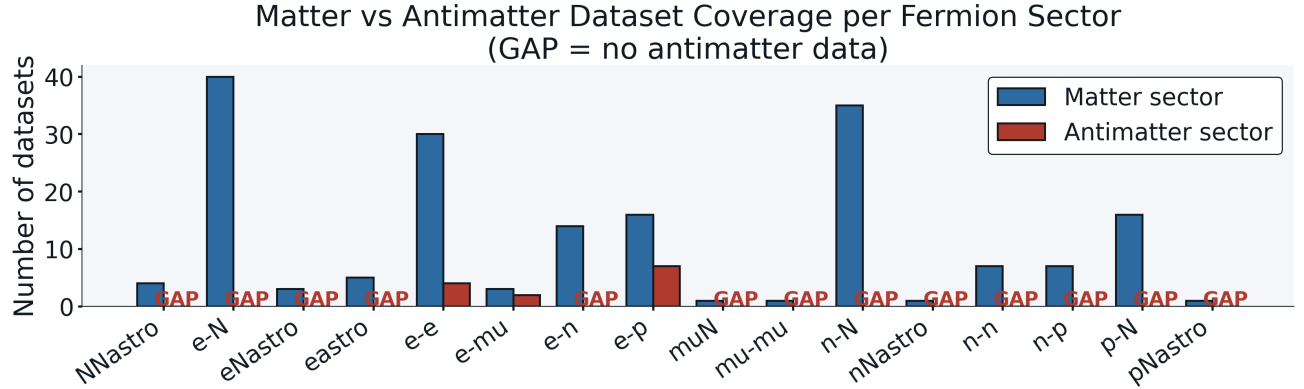


FIG. 5. Matter versus antimatter dataset coverage per fermion sector. Sectors marked GAP have no antimatter-sector data at all.

not a parsing gap that a naming convention could close, since no single V_n describes an envelope over several experiments, and they would have to be decomposed into constituent bounds before entering a per-potential analysis. Two are contributor-template placeholders that were previously being read as V_1 and V_2 data on the strength of a trailing digit in their filenames. The remaining one is a $g_p g_p$ astrophysical e - N bound—a real measurement

whose potential is unassigned upstream, and for which assigning V_{1a} from sibling filenames would be inference rather than evidence.

Second, six entries under V_3 , V_{9+10} and V_{15} report interaction ranges that are physically implausible, reaching $\lambda \sim 10^{52}$ m against an observable universe of order 10^{26} m, almost certainly from a mis-detected unit token. Two of the fifteen matched pairs draw on an af-

fected dataset: the $g_{AGA} \cdot V_3 \cdot e-e^+$ pair on the matter side and the $g_{AGA} \cdot V_3 \cdot e-\bar{p}$ pair on the antimatter side. Neither result is compromised, because the asymmetry is evaluated only on the overlapping range of the two curves and in both cases the implausible tail lies outside it: the ranges used are $[2.5 \times 10^{-12}, 1.6 \times 10^{-5}]$ m and $[2.0 \times 10^{-11}, 1.3 \times 10^{-8}]$ m respectively. A broader coverage claim drawn from the full database without this cleanup would nonetheless not be reliable.

VIII. CONCLUSIONS

We have constructed a systematic dictionary between the SME fermion-sector coefficients b_μ , $H_{\mu\nu}$ and $d_{\mu\nu}$ and the Dobrescu–Mocioiu potential basis by explicit Foldy–Wouthuysen reduction, summarized in Table I. All three reductions are complete and agree with established results. The apparent sign discrepancy in the momentum-linear d_{ij} and d_{00} components against Ref. [5] is traced to the momentum-index convention of that reference, fixed by the free-particle term of the corresponding derivation [6], and is not a feature of the reduction.

Using this dictionary we defined a CPT asymmetry parameter and evaluated it against 247 standardized experimental constraint datasets. Fifteen matter–antimatter pairs could be compared, and all show statistically significant asymmetry after autocorrelation correction. We showed, however, that this significance cannot be read as evidence of CPT violation: an asymmetry built from one-sided upper bounds saturates whenever the two ex-

periments differ in sensitivity, and a CPT-even channel in our sample produces an asymmetry comparable to the CPT-odd channels. The corollary is a concrete experimental design criterion—sensitivity matching, not absolute tightness, is what makes an antimatter-sector bound informative for a CPT test.

The substantive result is what this exposes about the data. Only 19 of 247 analyzed datasets involve an antimatter sector; seven of twelve classified potentials and six of eleven fermion-sector pairs have zero antimatter-sector coverage despite substantial matter-sector data. The binding constraint on testing CPT symmetry through exotic spin-dependent interactions is therefore not matter-sector precision, nor statistical power, but the near-total absence of comparably precise antimatter-sector measurements in exactly the channels where matter-sector data is most abundant. We offer the mapping table, the compiled database and the analysis pipeline as an extensible resource for closing that gap, designed so that future antimatter-sector measurements are incorporated as they are published.

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